

SOFTWARE ENGINEERING

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

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Problem Statement #36: Restaurant Table Booking & Pre-Ordering App

1. Scenario Review

System: Restaurant Table Booking & Pre-Ordering App

Functional Requirements:

- FR-001: Select an available table from the interactive 2D floor plan and link pre-ordered menu items to the reservation.
- FR-002: Select menu items and add them to the reservation as a pre-order.
- FR-003: Select a reservation date and dining time for an available table.
- FR-004: Calculate the estimated kitchen preparation time for pre-ordered menu items.
- FR-005: Allow the restaurant manager to view and update restaurant table status.

Non-Functional Requirements:

- NFR-001: Table status changes (Reserved, Seated, Billed) must synchronize across all manager floor terminals in under 500 ms.
- NFR-002: The system shall protect diner reservation and pre-order information from unauthorized access or modification.

Main Challenges

- **Performance** — Table status changes must synchronize across manager floor terminals in under 500 ms.
- **Usability** — Diners should be able to select tables using the interactive 2D floor plan.
- **Security** — Reservation and pre-order information must be protected from unauthorized access or modification.
- **Overbooking Prevention** — The system must prevent reservations during overlapping dining periods.
- **Data Consistency** — Table statuses must remain consistent across manager floor terminals.

2. Architectural Style Analysis

2.1 Layered Architecture

Advantages:

- Provides clear separation between presentation, business logic and data.
- Easy to understand and maintain.

Disadvantages:

- Can introduce performance overhead between layers.
- Individual components are difficult to scale independently.

Suitability for the Restaurant Table Booking & Pre-Ordering App:

Moderately suitable because the application has separate presentation, business and data requirements. However, additional layer communication may affect the required fast table-status synchronization.

2.2 Microservices Architecture

Advantages:

- Allows functionalities such as reservation, food pre-ordering and kitchen preparation to be developed and scaled independently.
- Provides fault isolation between services.

Disadvantages:

- More complex to manage and deploy.
- Communication between services can introduce network latency and data consistency challenges.

Suitability for the Restaurant Table Booking & Pre-Ordering App:

Highly suitable because the application contains distinct functionalities such as table reservation, food pre-ordering, kitchen preparation and table-status management that can operate as separate services.

2.3 Client-Server Architecture

Advantages:

- Provides centralized control of application data.
- Makes maintaining consistent reservation information easier.

Disadvantages:

- The central server can become a scalability bottleneck.
- The server can become a single point of failure.

Suitability for the Restaurant Table Booking & Pre-Ordering App:

Suitable for a smaller deployment, but less suitable when supporting multiple diners and manager terminals simultaneously.

Architecture	Suitability
Layered Architecture	Moderately Suitable
Microservices Architecture	Highly Suitable
Client-Server Architecture	Suitable for smaller deployments

3.1 Architecture Selection

Selected Architecture: Microservices Architecture

We chose **Microservices Architecture** for the Restaurant Table Booking & Pre-Ordering App.

Reason 1: The application contains distinct functionalities such as table reservation, food pre-ordering, kitchen preparation and table-status management. These functionalities can be separated into independent services.

Reason 2: Microservices supports independent scaling and fault isolation, which is useful for handling multiple diners and restaurant manager terminals.

Security Advantage: Separate services can enforce controlled access to reservation and pre-order information, helping prevent unauthorized access or modification.

Performance Benefit: Individual services can be independently scaled to handle high-demand operations and support fast table-status updates.

3.2 Component Identification

1. Diner Interface Component

Provides the diner-facing interface for viewing the floor plan, selecting tables, choosing reservation times and selecting pre-order items.

2. Reservation Management Component

Manages table availability, reservation creation and prevention of overlapping reservations.

3. Food Pre-Order Management Component

Manages menu selection, quantities and linking pre-ordered food items to reservations.

4. Kitchen Preparation Service Component

Processes pre-order information and calculates the estimated kitchen preparation timeline.

5. Table Status Management Component

Manages and synchronizes table statuses such as Reserved, Seated and Billed.

4. Component Diagram

Figure 1: UML Component Diagram for the Restaurant Table Booking & Pre-Ordering App

